Data Analysis
AMS315

Chapter 6

The Normal Distribution and Other Continuous Distributions

Learning Objectives

In this chapter, you learn:

- To compute probabilities from the normal distribution
- To use the normal probability plot to determine whether a set of data is approximately normally distributed
- To compute probabilities from the uniform distribution
- To compute probabilities from the exponential distribution
- To compute probabilities from the normal distribution to approximate probabilities from the binomial distribution

Probability Distributions

Probability Distributions

Ch. 5

Discrete Probability Distributions

- Binomial
- Poisson
- Hypergeometric

Continuous Probability Distributions

Ch. 6

- Normal
- Uniform
- Exponential

Continuous Probability Distributions

- A continuous random variable is a variable that can assume any value on a continuum (can assume an uncountable number of values)
  - thickness of an item
  - time required to complete a task
  - temperature of a solution
  - height, in inches
- These can potentially take on any value, depending only on the ability to measure accurately.

The Normal Distribution

Probability Distributions

Continuous Probability Distributions

- Normal
- Uniform
- Exponential

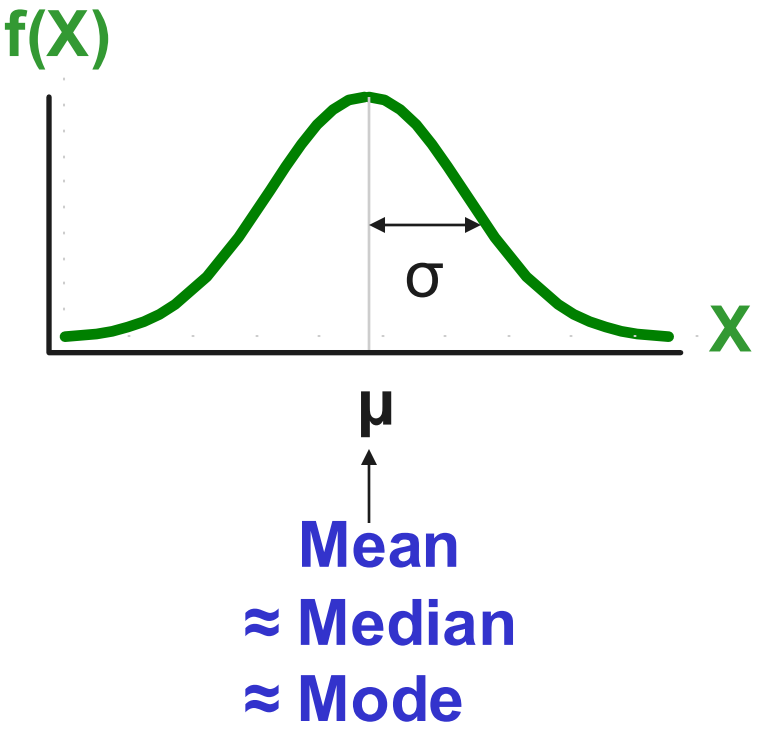
The Normal Distribution

- Bell Shaped
- Symmetrical
- Mean, Median and Mode are approx. Equal

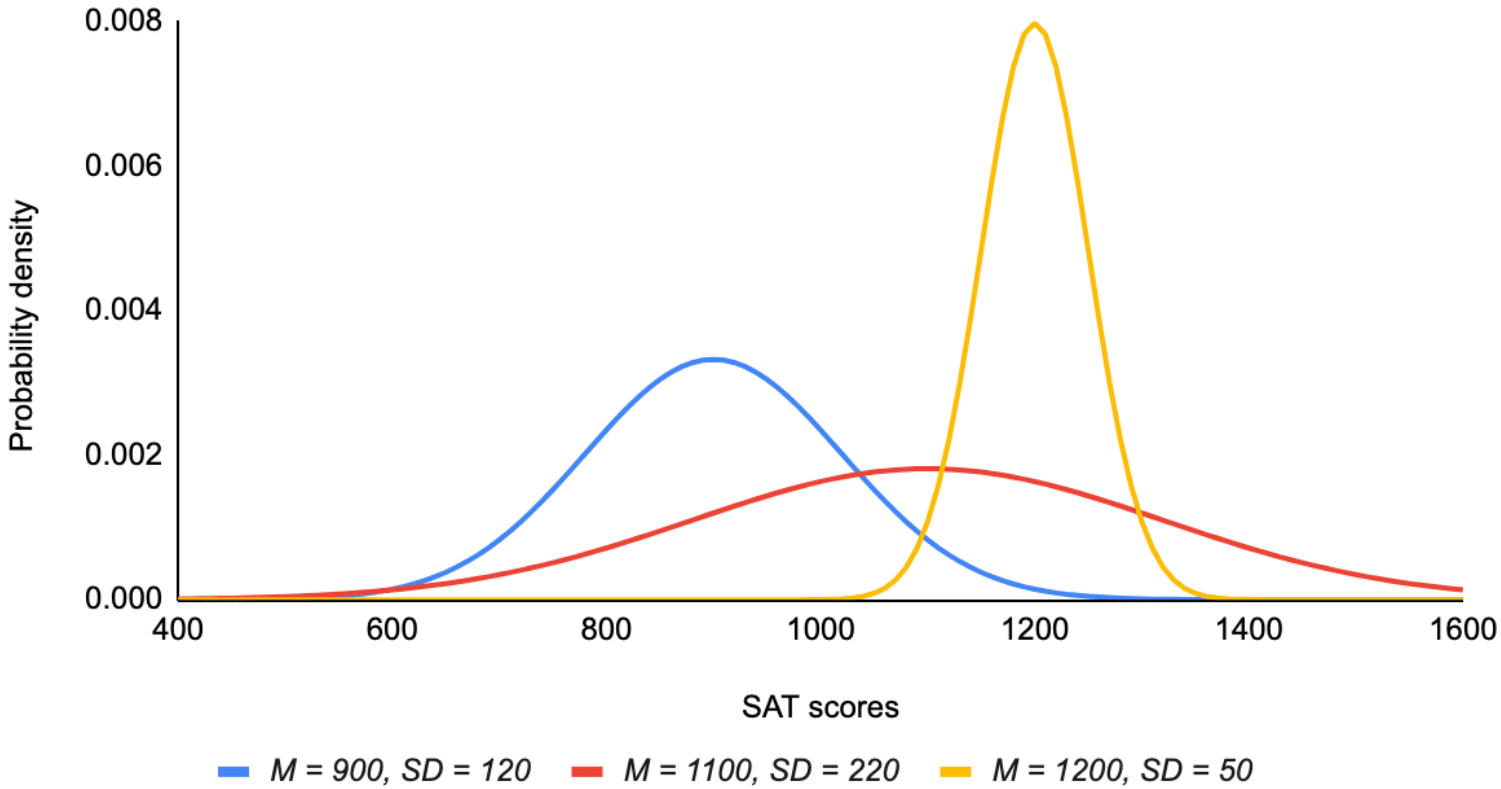
Location is determined by the mean, μ

Spread is determined by the standard deviation, σ

The random variable has an infinite theoretical range:
$+\infty$ to $-\infty$



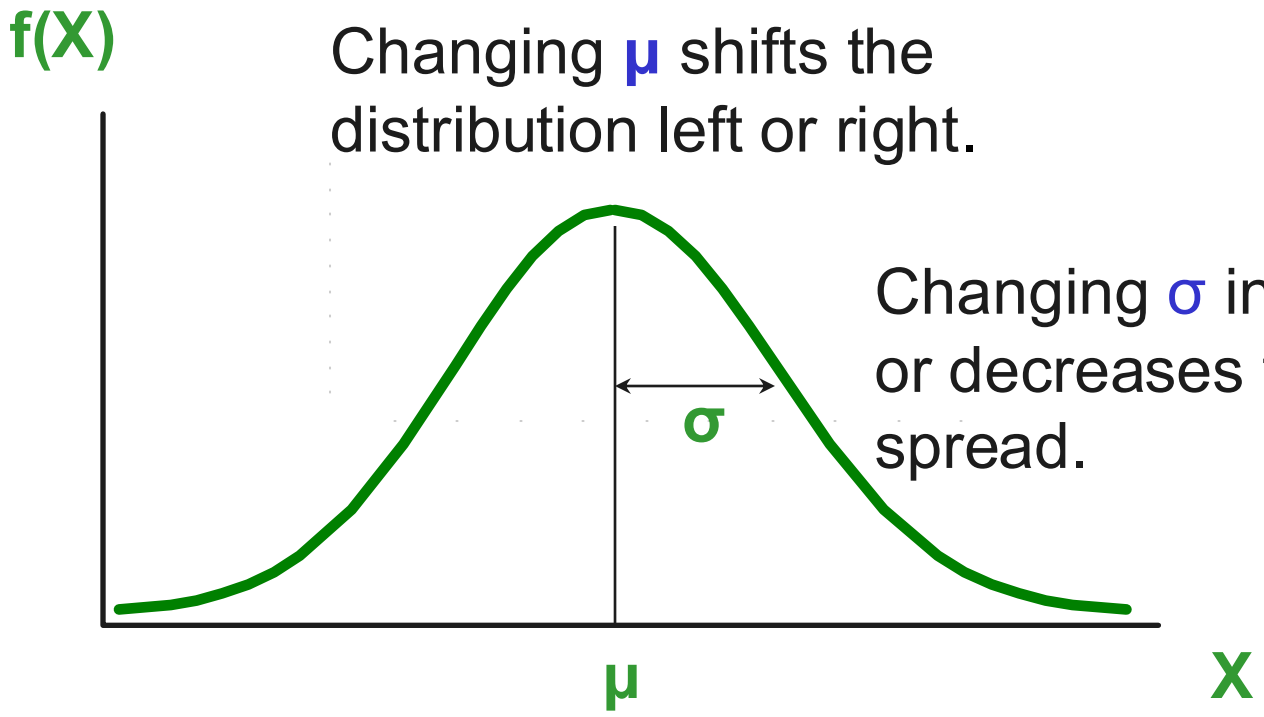
Many Normal Distributions



By varying the parameters μ and σ, we obtain different normal distributions

The Normal Distribution Shape

Changing μ shifts the distribution left or right.



Changing σ increases or decreases the spread.

The Normal Probability Density Function

- The formula for the normal probability density function is

$$f(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(1/2)[(X-\mu)/\sigma]^2}$$

Where e = the mathematical constant approximated by 2.71828

π = the mathematical constant approximated by 3.14159

μ = the population mean

σ = the population standard deviation

X = any value of the continuous variable

The Standardized Normal

- Any normal distribution (with any mean and standard deviation combination) can be transformed into the standardized normal distribution (Z)

- Need to transform X units into Z units

Translation to the Standardized Normal Distribution

- Translate from X to the standardized normal (the "Z" distribution) by subtracting the mean of X and dividing by its standard deviation:

$$Z = \frac{X - \mu}{\sigma}$$

The Z distribution always has mean = 0 and standard deviation = 1

The Standardized Normal Probability Density Function

- The formula for the standardized normal probability density function is

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-(1/2)Z^2}$$

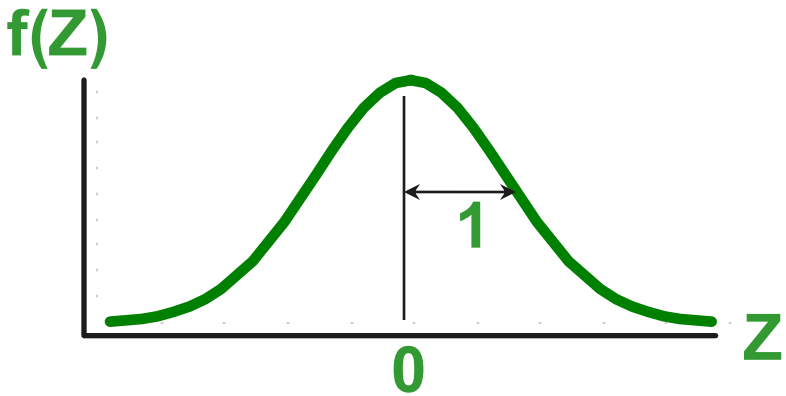
Where e = the mathematical constant approximated by 2.71828

π = the mathematical constant approximated by 3.14159

Z = any value of the standardized normal distribution

The Standardized Normal Distribution

- Also known as the “Z” distribution
- Mean is 0
- Standard Deviation is 1



Values above the mean have positive Z-values, values below the mean have negative Z-values

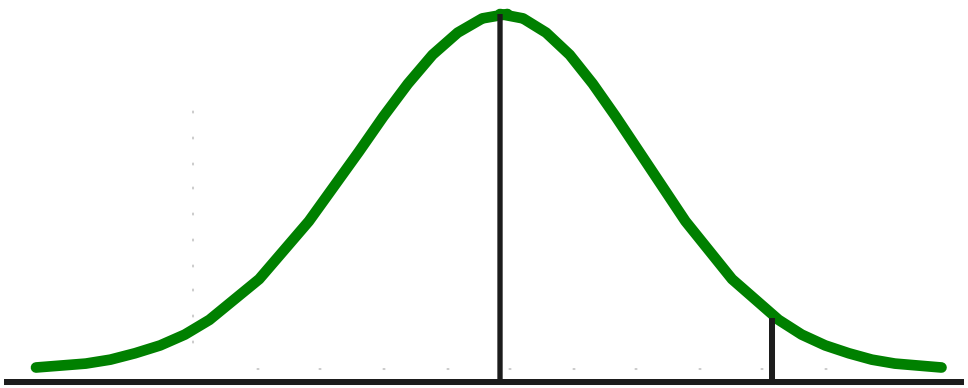
Example

- If X is distributed normally with mean of 100 and standard deviation of 50, the Z value for $X = 200$ is

$$Z = \frac{X - \mu}{\sigma} = \frac{200 - 100}{50} = 2.0$$

- This says that $X = 200$ is two standard deviations (2 increments of 50 units) above the mean of 100.

Comparing X and Z units

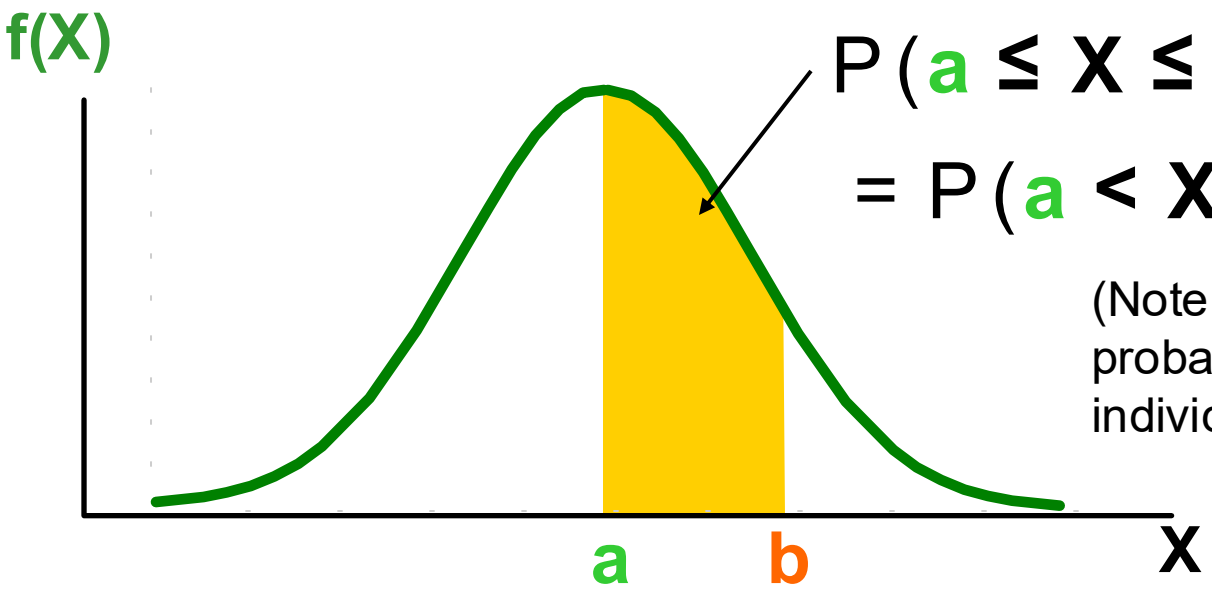


100	**200**	**X**	(μ = 100, σ = 50)
0	**2.0**	**Z**	(μ = 0, σ = 1)

Note that the distribution is the same, only the scale has changed. We can express the problem in original units (X) or in standardized units (Z)

Finding Normal Probabilities

Probability is measured by the area under the curve



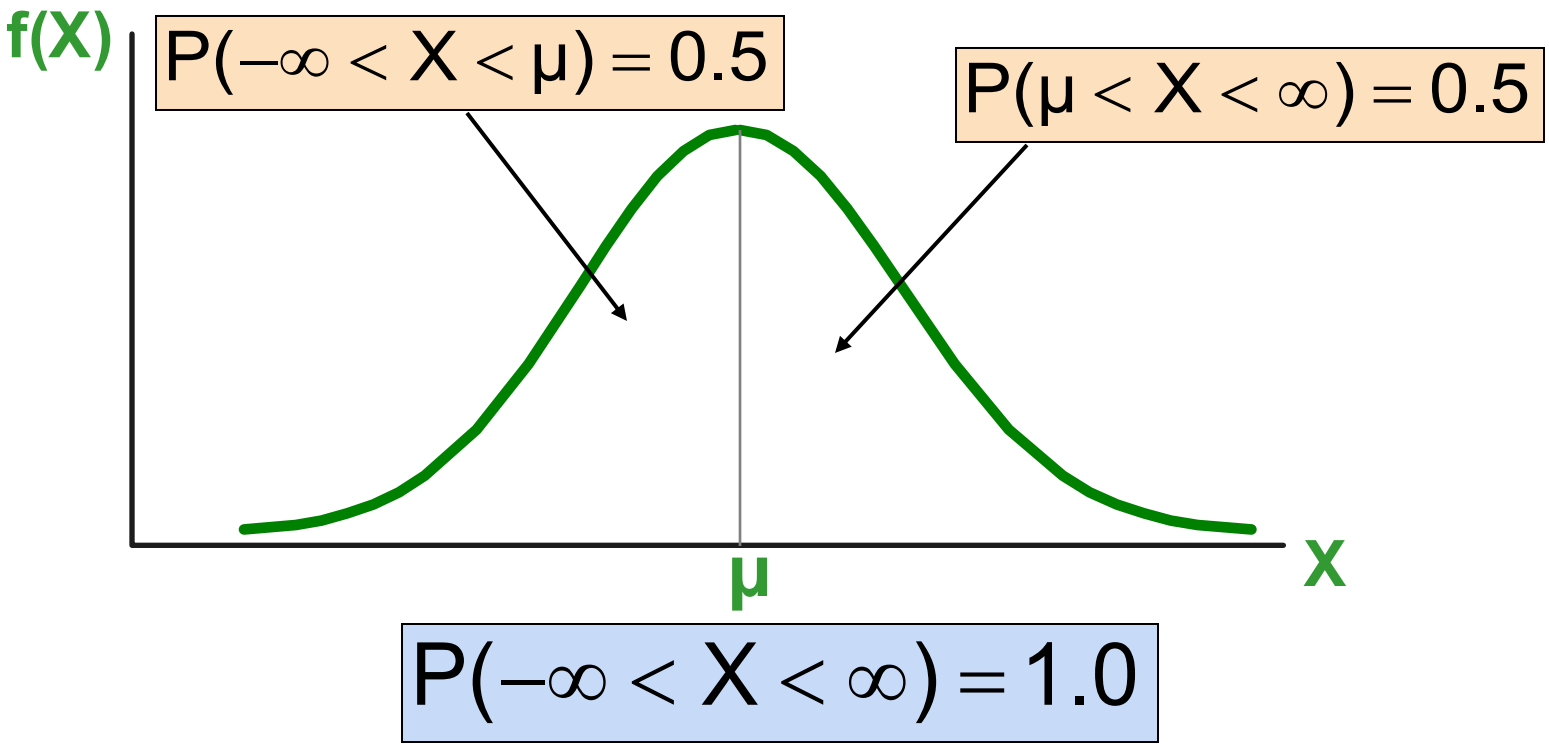
$$P(a \le X \le b)$$

$$= P(a < X < b)$$

(Note that the probability of any individual value is zero)

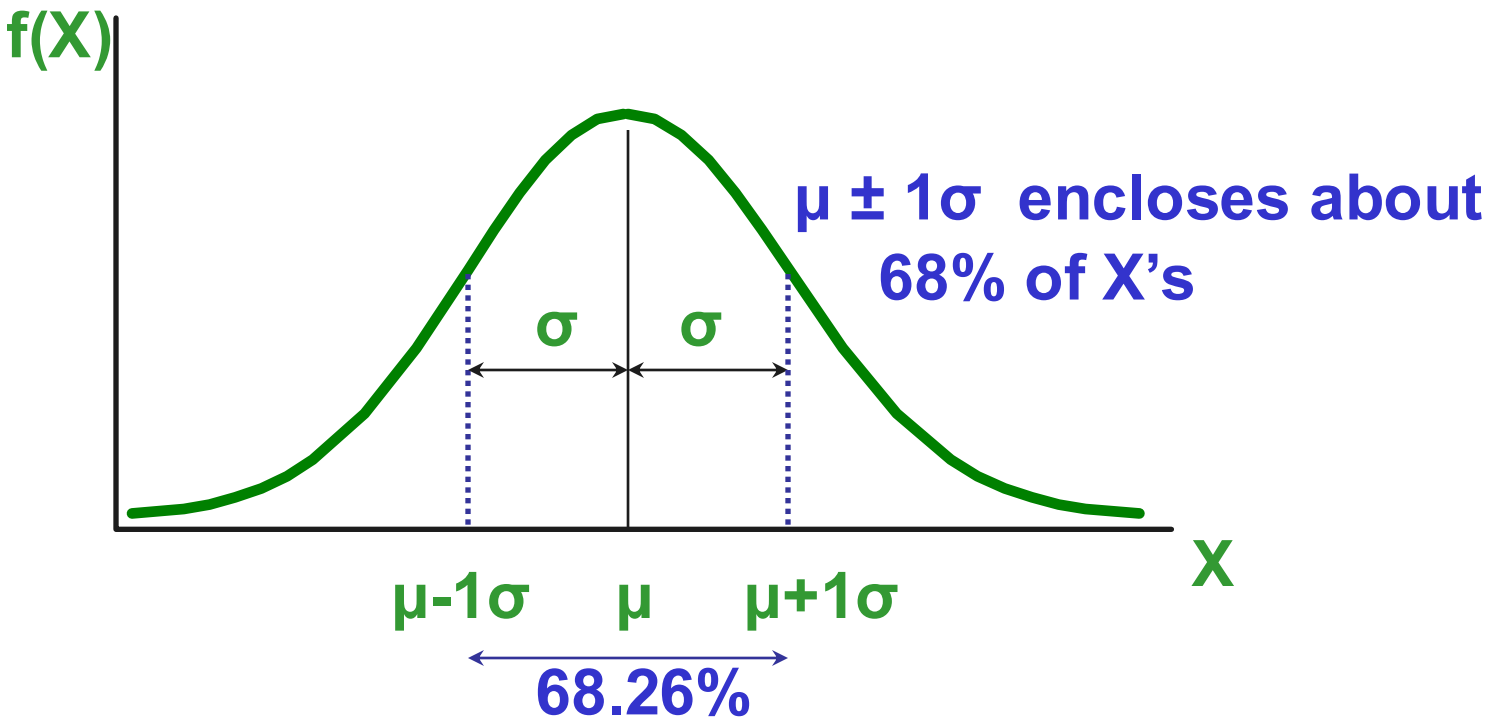
Probability as Area Under the Curve

The total area under the curve is 1.0, and the curve is symmetric, so half is above the mean, half is below



Empirical Rules

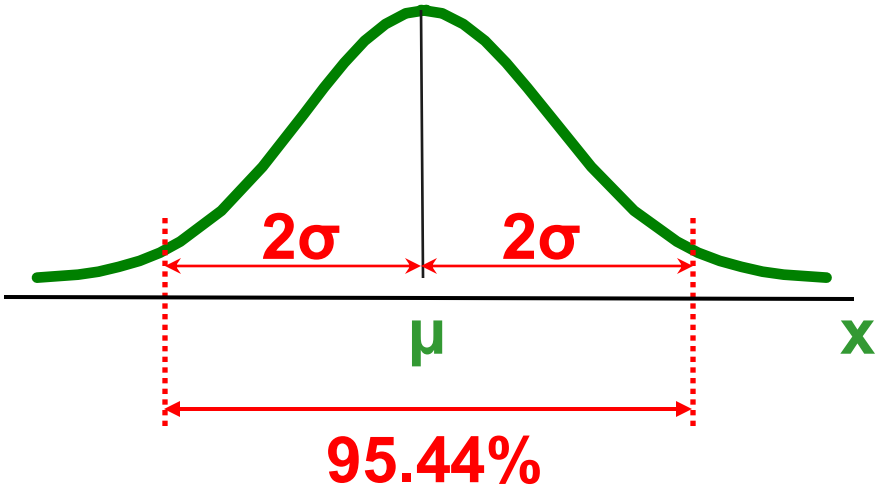
What can we say about the distribution of values around the mean? There are some general rules:

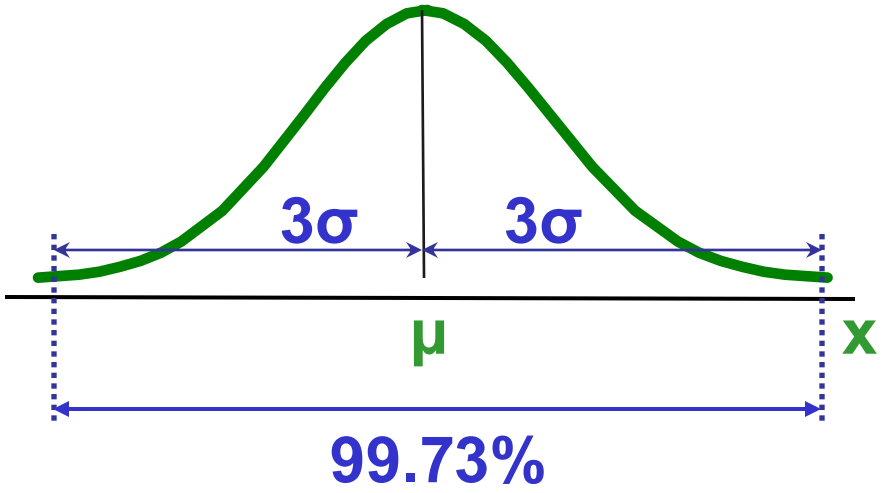


The Empirical Rule

(continued)

- $\mu \pm 2\sigma$ covers about 95% of X's
- $\mu \pm 3\sigma$ covers about 99.7% of X's



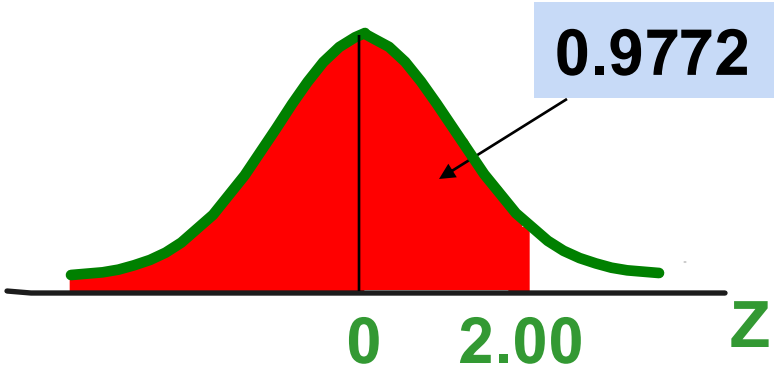


The Standardized Normal Table

The Cumulative Standardized Normal table gives the probability less than a desired value for Z (i.e., from negative infinity to Z)

Example:

$P(Z < 2.00) = 0.9772$



The Standardized Normal Table

(continued)

The column gives the value of Z to the second decimal point

The row shows the value of Z to the first decimal point

Z	0.00	0.01	0.02 …
0.0			
0.1			
⋮			
2.0	**.9772**		

The value within the table gives the probability from Z = – ∞ up to the desired Z value

P(Z < 2.00) = 0.9772

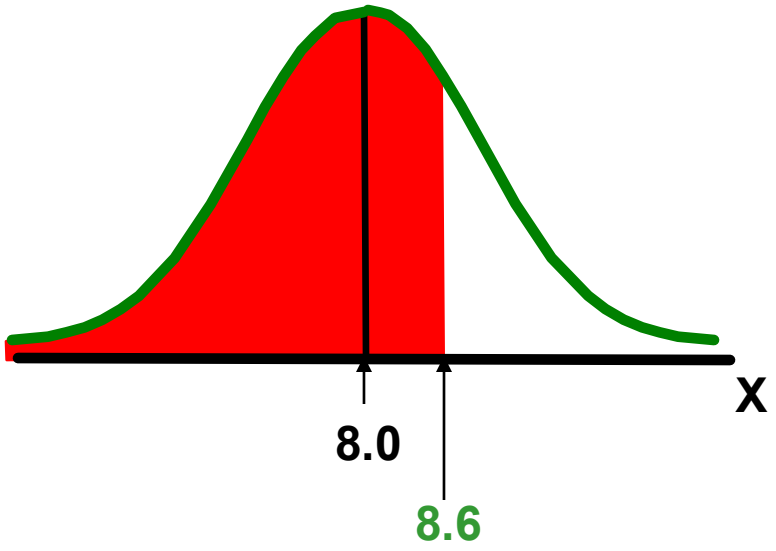
General Procedure for Finding Probabilities

To find $P(a < X < b)$ when X is distributed normally:

- Draw the normal curve for the problem in terms of X
- Translate X-values to Z-values
- Use the Standardized Normal Table

Finding Normal Probabilities

- Suppose X is normal with mean 8.0 and standard deviation 5.0
- Find $P(X < 8.6)$

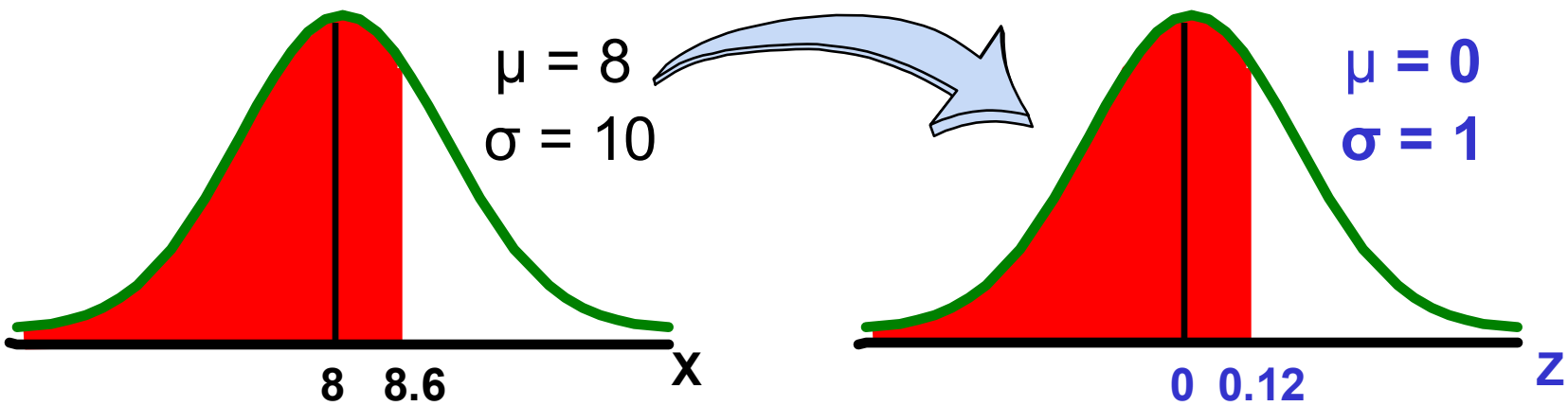


Finding Normal Probabilities

(continued)

- Suppose X is normal with mean 8.0 and standard deviation 5.0. Find P(X < 8.6)

$$Z = \frac{X - \mu}{\sigma} = \frac{8.6 - 8.0}{5.0} = 0.12$$



P(X < 8.6)

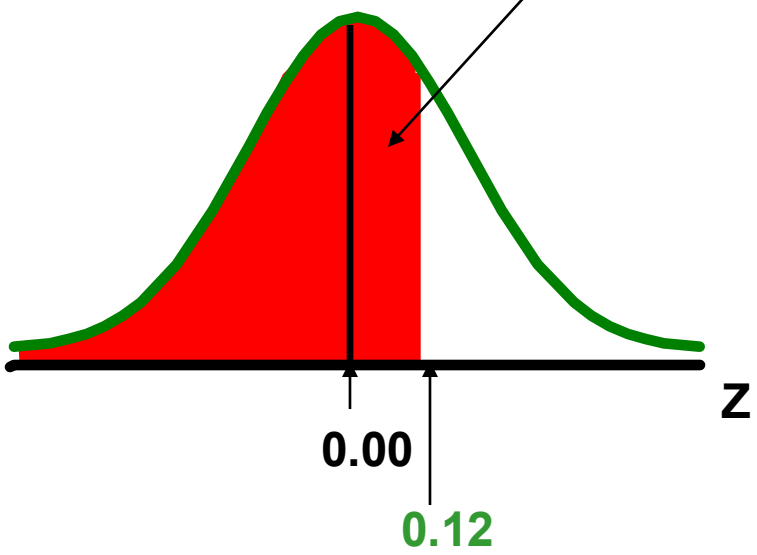
P(Z < 0.12)

Solution: Finding P(Z < 0.12)

Standardized Normal Probability Table (Portion)

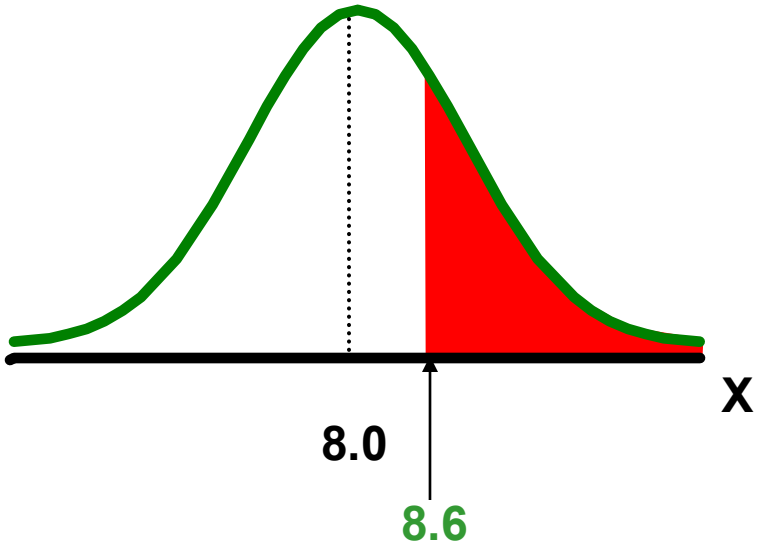
Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255

P(X < 8.6)
= P(Z < 0.12)
.5478



Upper Tail Probabilities

- Suppose X is normal with mean 8.0 and standard deviation 5.0.
- Now Find P(X > 8.6)



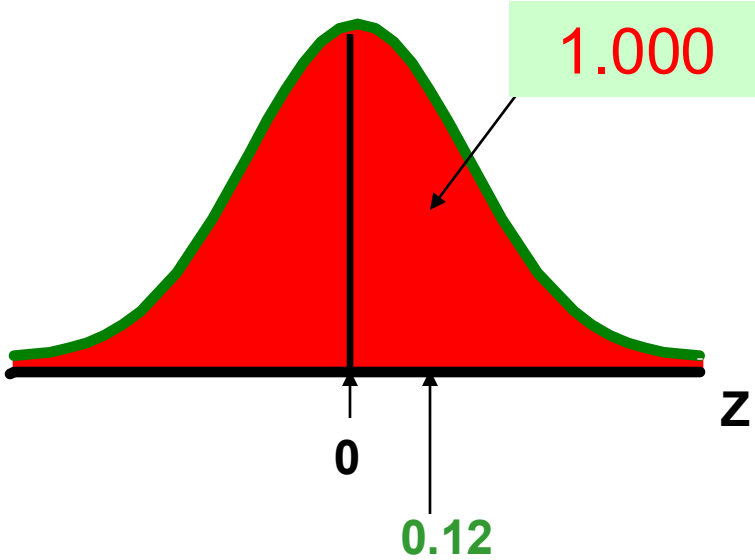
Upper Tail Probabilities

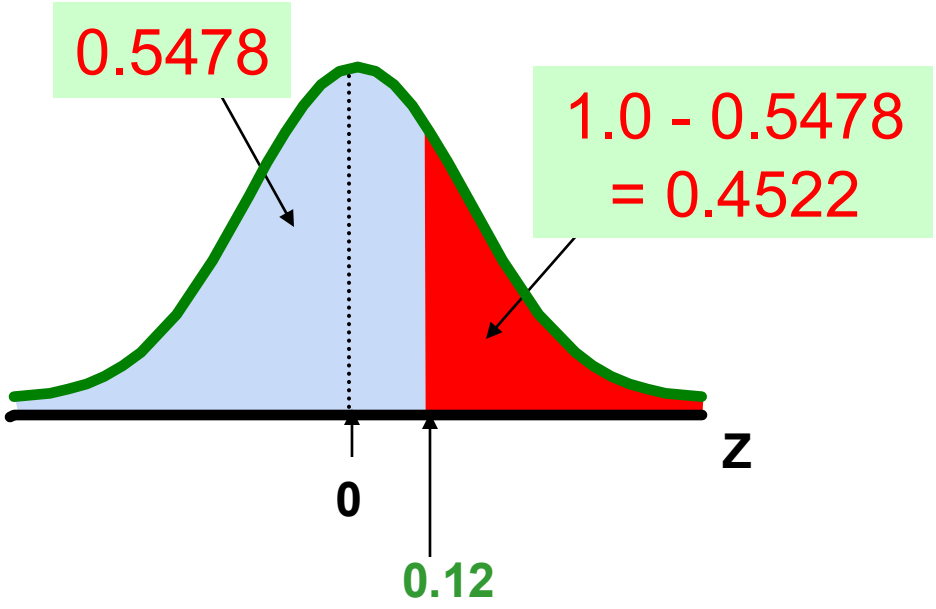
(continued)

- Now Find $P(X > 8.6)$…

$$P(X > 8.6) = P(Z > 0.12) = 1.0 - P(Z \le 0.12)$$

$$= 1.0 - 0.5478 = \boxed{0.4522}$$





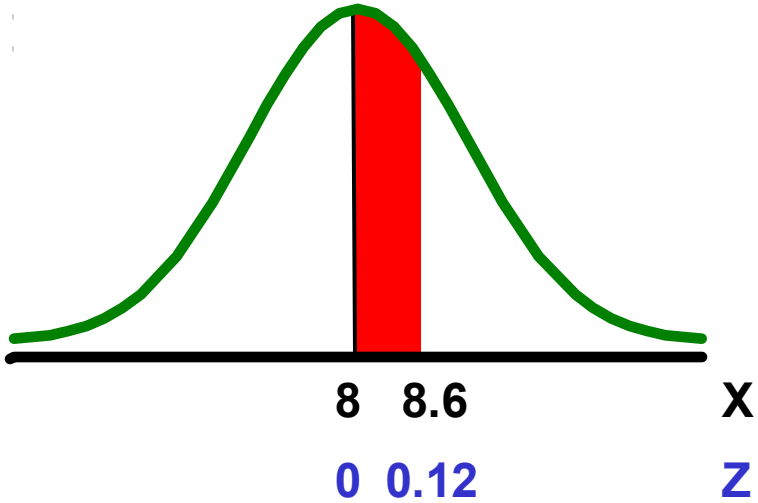
Probability Between Two Values

- Suppose X is normal with mean 8.0 and standard deviation 5.0. Find P(8 < X < 8.6)

Calculate Z-values:

$$Z = \frac{X - \mu}{\sigma} = \frac{8 - 8}{5} = 0$$

$$Z = \frac{X - \mu}{\sigma} = \frac{8.6 - 8}{5} = 0.12$$



P(8 < X < 8.6)

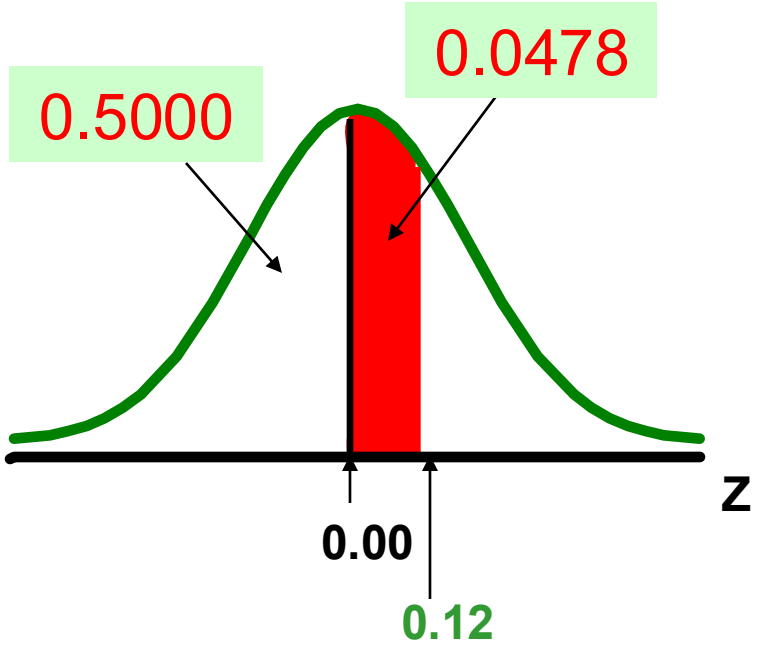
= P(0 < Z < 0.12)

Solution: Finding P(0 < Z < 0.12)

Standardized Normal Probability Table (Portion)

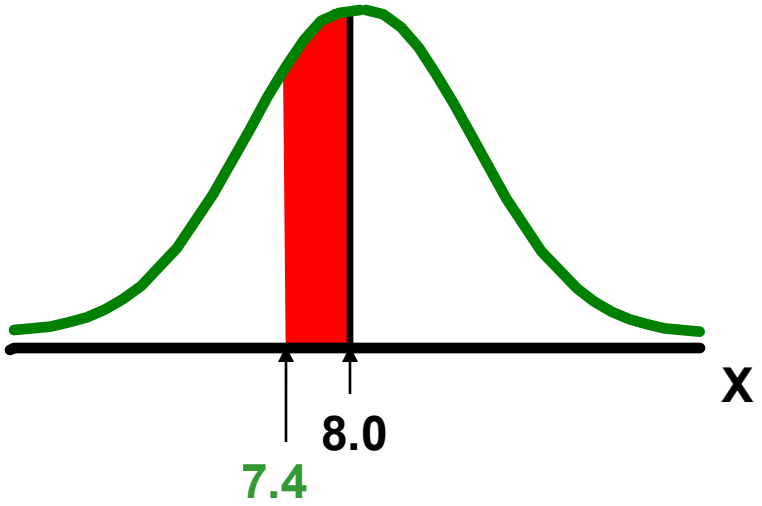
Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255

$P(8 < X < 8.6)$
$= P(0 < Z < 0.12)$
$= P(Z < 0.12) - P(Z \leq 0)$
$= 0.5478 - .5000 = 0.0478$



Probabilities in the Lower Tail

- Suppose X is normal with mean 8.0 and standard deviation 5.0.
- Now Find $P(7.4 < X < 8)$



Probabilities in the Lower Tail

(continued)

Now Find P(7.4 < X < 8)…

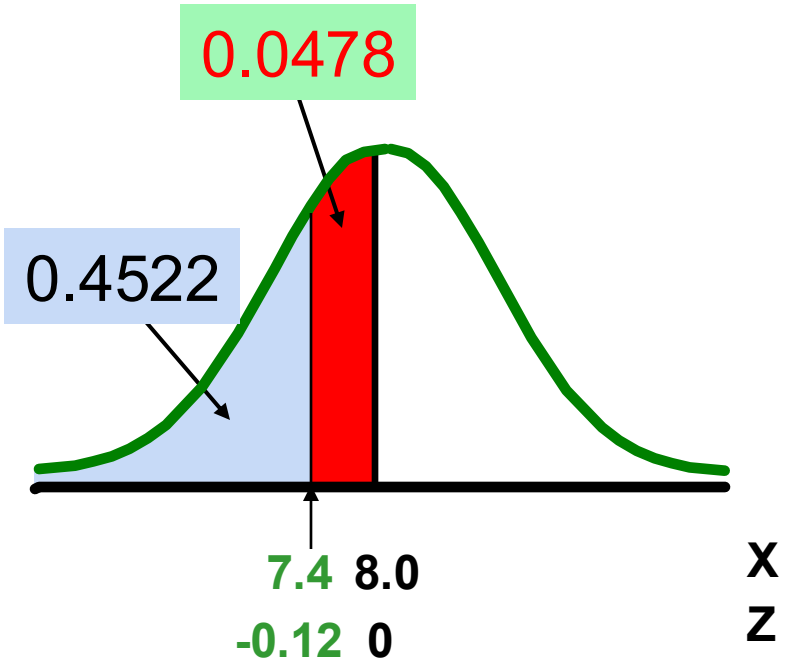
$P(7.4 < X < 8)$

$= P(-0.12 < Z < 0)$

$= P(Z < 0) – P(Z \leq -0.12)$

$= 0.5000 - 0.4522 = \boxed{0.0478}$

The Normal distribution is symmetric, so this probability is the same as $P(0 < Z < 0.12)$



Finding the X value for a Known Probability

- Steps to find the X value for a known probability:
  1. Find the Z value for the known probability
  2. Convert to X units using the formula:

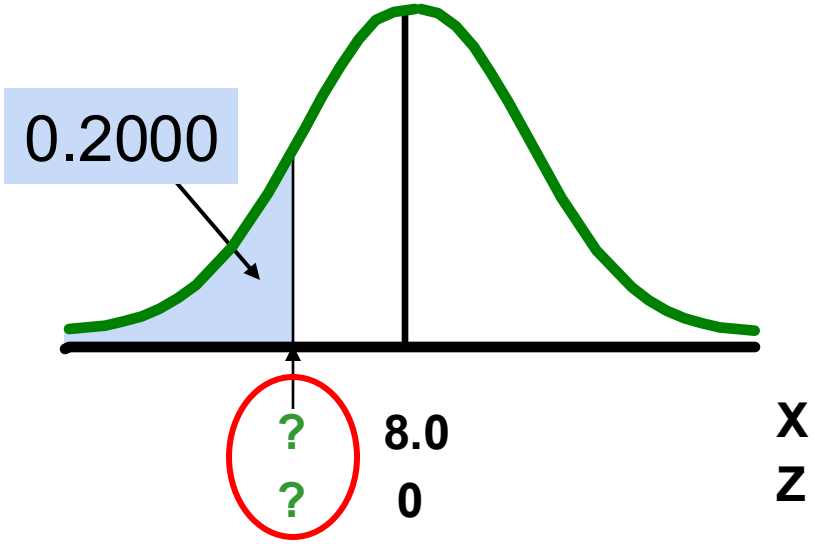
$$X = \mu + Z\sigma$$

Finding the X value for a Known Probability

(continued)

Example:

- Suppose X is normal with mean 8.0 and standard deviation 5.0.
- Now find the X value so that only 20% of all values are below this X



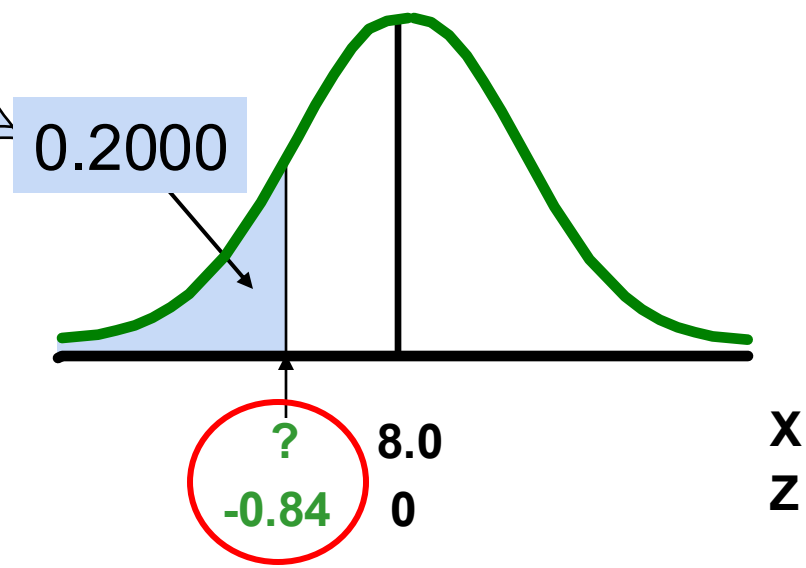
Find the Z value for 20% in the Lower Tail

1. Find the Z value for the known probability

Standardized Normal Probability Table (Portion)

Z	…	.03	.04	.05
-0.9	…	.1762	.1736	.1711
-0.8	…	.2033	.2005	.1977
-0.7	…	.2327	.2296	.2266

- 20% area in the lower tail is consistent with a Z value of -0.84



Finding the X value

2. Convert to X units using the formula:

$$X = \mu + Z\sigma$$

$$= 8.0 + (-0.84)5.0$$

$$= 3.80$$

So 20% of the values from a distribution with mean 8.0 and standard deviation 5.0 are less than 3.80

Evaluating Normality

- Not all continuous random variables are normally distributed
- It is important to evaluate how well the data set is approximated by a normal distribution

Evaluating Normality

(continued)

- Construct charts or graphs
  - For small- or moderate-sized data sets, do stem-and-leaf display and box-and-whisker plot look symmetric?
  - For large data sets, does the histogram or polygon appear bell-shaped?
- Compute descriptive summary measures
  - Do the mean, median and mode have similar values?
  - Is the interquartile range approximately 1.33 σ?
  - Is the range approximately 6 σ?

Assessing Normality

(continued)

- Observe the distribution of the data set
  - Do approximately 2/3 of the observations lie within mean $\pm$ 1 standard deviation?
  - Do approximately 80% of the observations lie within mean $\pm$ 1.28 standard deviations?
  - Do approximately 95% of the observations lie within mean $\pm$ 2 standard deviations?
- Evaluate normal probability plot
  - Is the normal probability plot approximately linear with positive slope?

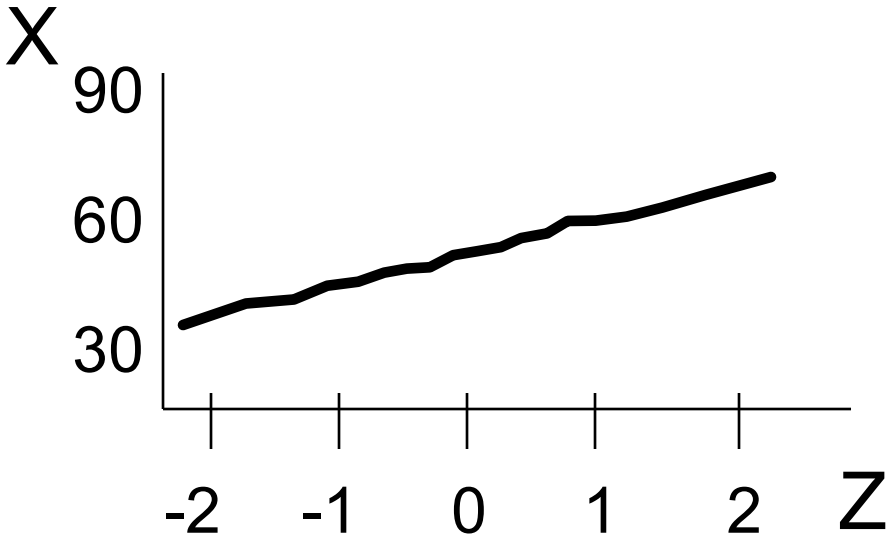
The Normal Probability Plot

- Normal probability plot
  - Arrange data into ordered array
  - Find corresponding standardized normal quantile values
  - Plot the pairs of points with observed data values on the vertical axis and the standardized normal quantile values on the horizontal axis
  - Evaluate the plot for evidence of linearity

The Normal Probability Plot

(continued)

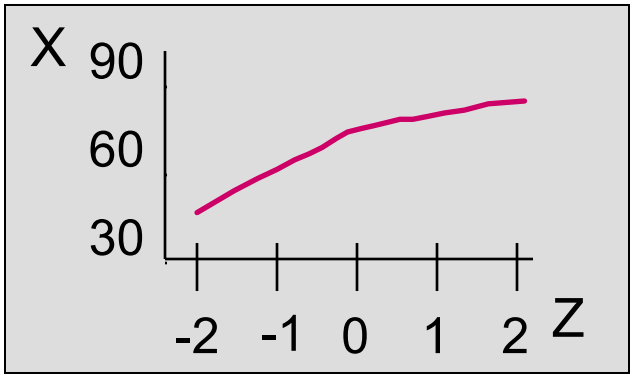
A normal probability plot for data from a normal distribution will be approximately linear:



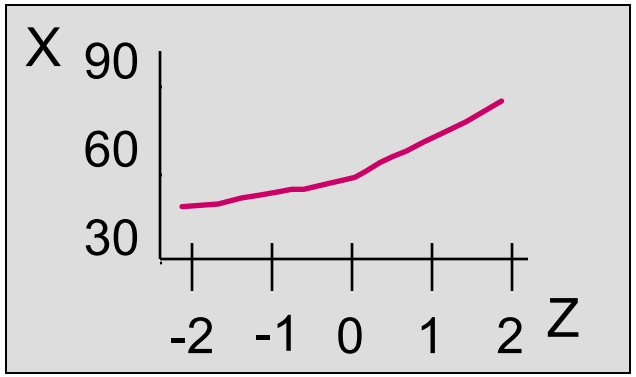
Normal Probability Plot

(continued)

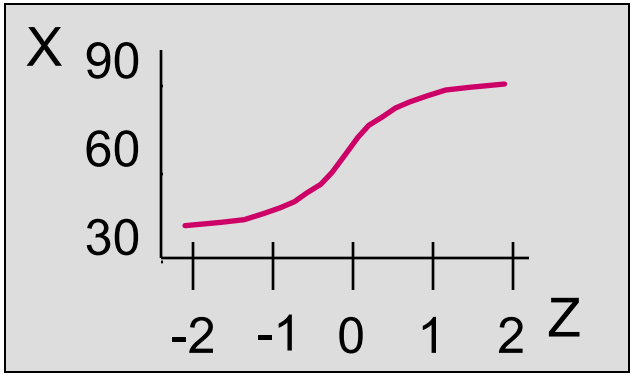
Left-Skewed



Right-Skewed

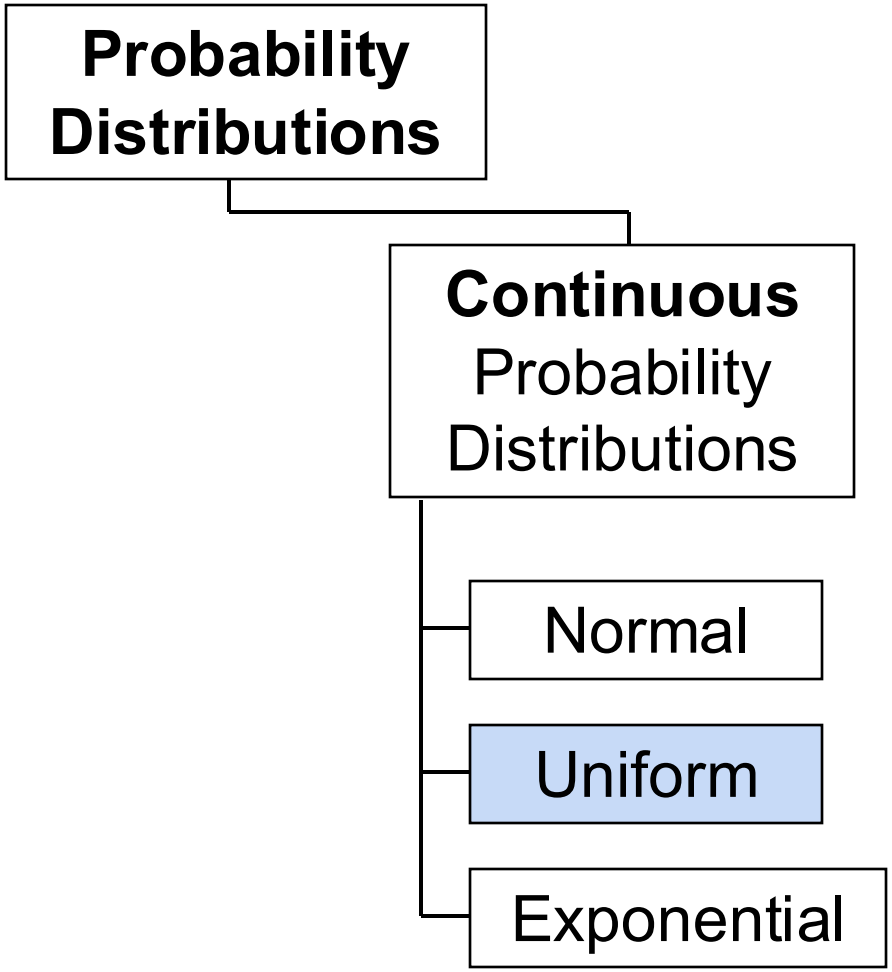


Rectangular



Nonlinear plots indicate a deviation from normality

The Uniform Distribution



The Uniform Distribution

- The uniform distribution is a probability distribution that has equal probabilities for all possible outcomes of the random variable

- Also called a rectangular distribution

The Uniform Distribution

(continued)

The Continuous Uniform Distribution:

$$f(X) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \le X \le b \\ 0 & \text{otherwise} \end{cases}$$

where

f(X) = value of the density function at any X value

a = minimum value of X

b = maximum value of X

Properties of the Uniform Distribution

- The mean of a uniform distribution is

$$\mu = \frac{a+b}{2}$$

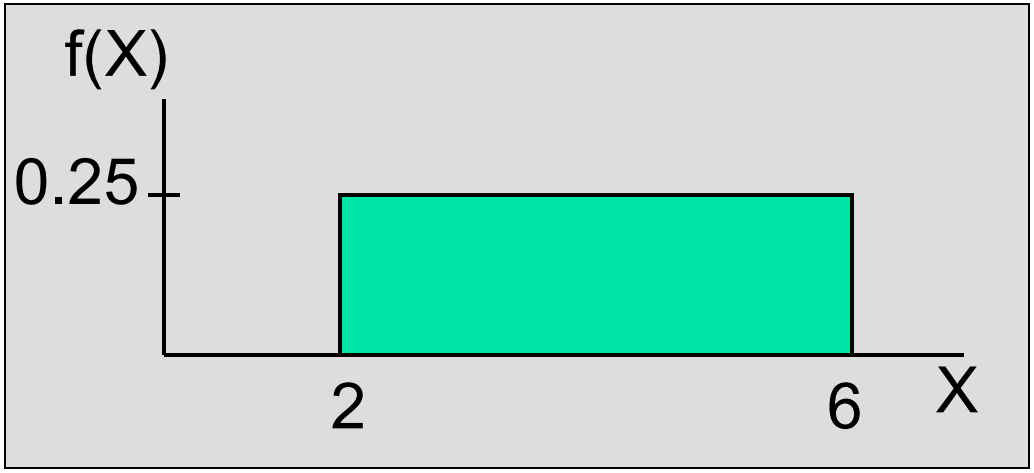
- The standard deviation is

$$\sigma = \sqrt{\frac{(b-a)^2}{12}}$$

Uniform Distribution Example

Example: Uniform probability distribution over the range $2 \le X \le 6$:

$$f(X) = \frac{1}{6 - 2} = 0.25 \quad \text{for } 2 \le X \le 6$$



$$\mu = \frac{a+b}{2} = \frac{2+6}{2} = 4$$

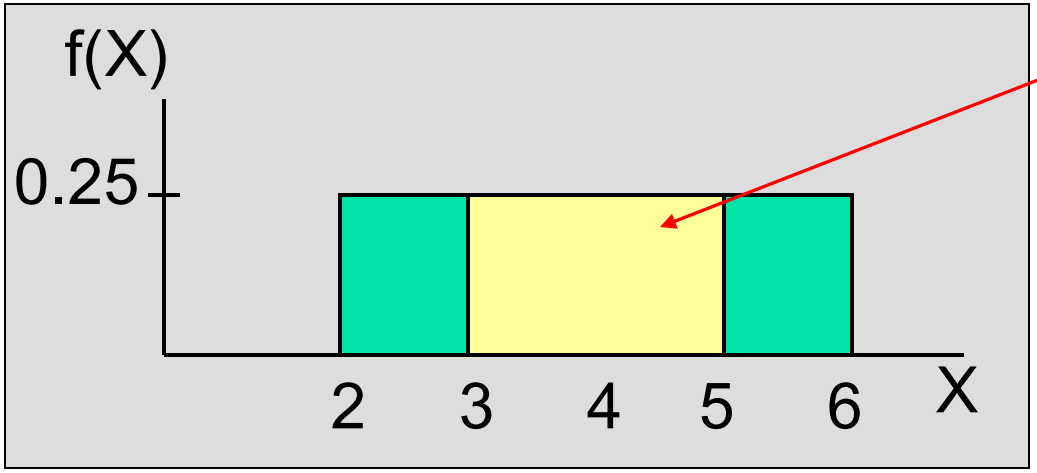
$$\sigma = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{(6-2)^2}{12}} = 1.1547$$

Uniform Distribution Example

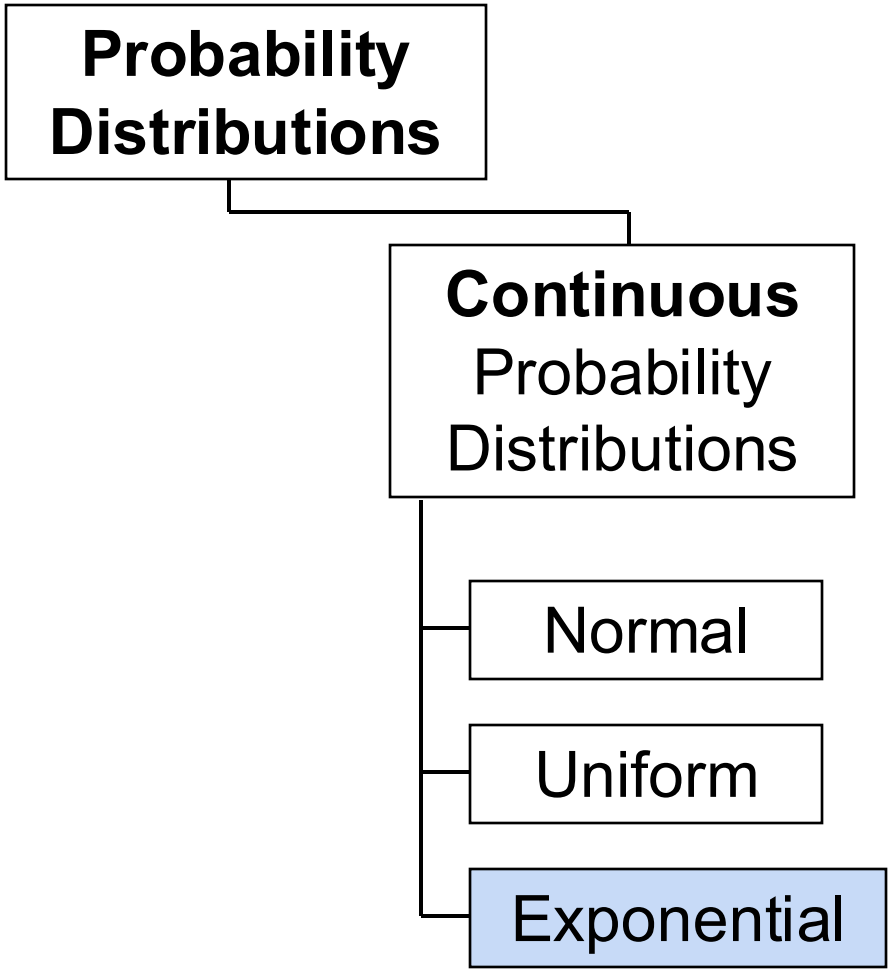
(continued)

Example: Using the uniform probability distribution to find $P(3 \le X \le 5)$:

$$P(3 \le X \le 5) = (\text{Base})(\text{Height}) = (2)(0.25) = 0.5$$



The Exponential Distribution



The Exponential Distribution

- Often used to model the length of time between two occurrences of an event (the time between arrivals)
  - Examples:
    - Time between trucks arriving at an unloading dock
    - Time between transactions at an ATM Machine
    - Time between phone calls to the main operator

The Exponential Distribution

- Defined by a single parameter, its mean λ (lambda)
- The probability that an arrival time is less than some specified time X is

$$P(\text{arrival time} < X) = 1 - e^{-\lambda X}$$

where e = mathematical constant approximated by 2.71828

λ = the population mean number of arrivals per unit

X = any value of continuous variable where $0 < X < \infty$

Exponential Distribution Example

Example: Customers arrive at the service counter at the rate of 15 per hour. What is the probability that the arrival time between consecutive customers is less than three minutes?

- The mean number of arrivals per hour is 15, so $\lambda = 15$
- Three minutes is 0.05 hours
- P(arrival time < .05) $= 1 - e^{-\lambda X} = 1 - e^{-(15)(0.05)} =$ $\boxed{0.5276}$
- So, there is a 52.76% probability that the arrival time between successive customers is less than three minutes

Normal Approximation to the Binomial Distribution

- The binomial distribution is a discrete distribution, but the normal is continuous
- To use the normal to approximate the binomial, accuracy is improved if you use a correction for continuity adjustment
- Example:
  - X is discrete in a binomial distribution, so $P(X = 4)$ can be approximated with a continuous normal distribution by finding

$$P(3.5 < X < 4.5)$$

Normal Approximation to the Binomial Distribution

(continued)

- The closer p is to 0.5, the better the normal approximation to the binomial
- The larger the sample size n, the better the normal approximation to the binomial
- General rule:
  - The normal distribution can be used to approximate the binomial distribution if

$$np \geq 5$$

and

$$n(1 - p) \geq 5$$

Normal Approximation to the Binomial Distribution

(continued)

- The mean and standard deviation of the binomial distribution are

$$\mu = np$$

$$\sigma = \sqrt{np(1-p)}$$

- Transform binomial to normal using the formula:

$$\boxed{Z = \frac{X-\mu}{\sigma} = \frac{X-np}{\sqrt{np(1-p)}}}$$

Using the Normal Approximation to the Binomial Distribution

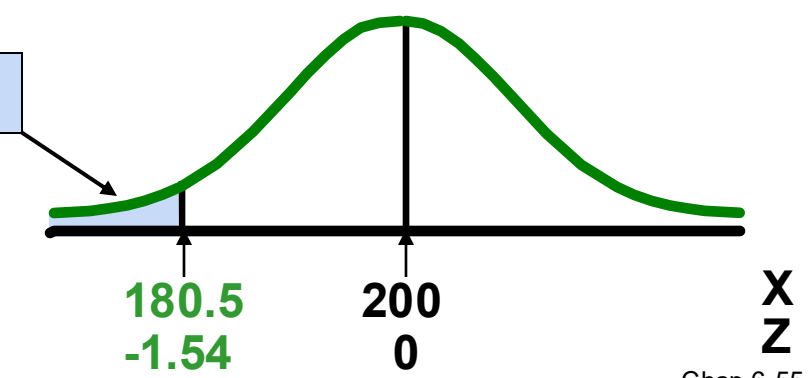
- If n = 1000 and p = 0.2, what is P(X ≤ 180)?
- Approximate P(X ≤ 180) using a continuity correction adjustment:

$$P(X \le 180.5)$$

- Transform to standardized normal:

$$Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{180.5 - (1000)(0.2)}{\sqrt{(1000)(0.2)(1-0.2)}} = -1.54$$

- So P(Z ≤ -1.54) = 0.0618



Using the Normal Approximation to the Binomial Distribution

- Class Tutorial:

- Use the normal approximation to the binomial distribution to find the probability of more than 35 successes when n = 40 and p = 0.75.

- Given that x is a continuous random variable with μ=20.5, σ=1.25, and n=49, find the probability that x is 10.

- Given the continuous probability distribution defined by f(x) =x/2 for the interval [0,2], find P(0.25≤ x ≤0.75).

Chapter Summary

- Presented key continuous distributions
  - normal, uniform, exponential
- Found probabilities using formulas and tables
- Recognized when to apply different distributions
- Applied distributions to decision problems